

```
1. package mongodb;
```

```
import java.sql.Connection;
```

```
import java.sql.DriverManager;
```

```
import java.sql.ResultSet;
```

```
import java.sql.Statement;
```

```
public class Testing {
```

```
    public static void main(String[] args){
```

```
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con =
```

```
            DriverManager.getConnection("jdbc:mysql://localhost:3306/exercise",
```

```
                "root","root123");
```

```
            Statement stmt = con.createStatement();
```

```
            ResultSet rs = stmt.executeQuery("select * from students");
```

```
            while(rs.next()) {
```

```
                System.out.println(rs.getInt(1)+" "+
```

```
                    rs.getString(2)+" "+rs.getInt(3));
```

```
            }
```

```
            con.close();
```

```
        }catch(Exception e){
```

```
            System.out.println("Sorry for the inconvenience happened..! "
```

```
                + "Check the inputs properly one more time");
```

```
        }
```

```
    }
```

}

```
Console x
<terminated> Testing [Java Application] C:\Users\rudra\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full
1 Divyansh 85
2 Rahul 78
3 Priya 92
4 Aman 67
5 Neha 88

");
ction

t * f

"+
Int(3

viner... happened...
```

2.CRUD_Test;

package CRUD;

import java.sql.Connection;

import java.sql.PreparedStatement;

import java.sql.ResultSet;

public class CRUD_Test {

public void create_data(String sl_no,String name,String email){

DB_Connection obj_DB_Connection=new DB_Connection();

Connection connection=obj_DB_Connection.get_connection();

PreparedStatement ps=null;

try{

String query="insert into student(sl_no,name,email) values
(?,?,?)";

ps=connection.prepareStatement(query);

ps.setString(1, sl_no);

ps.setString(2, name);

ps.setString(3, email);

System.**out**.println(ps);

ps.executeUpdate();

} catch (Exception e) {

System.**out**.println(e);

}

}

```

public void read_data(String sl_no){

    DB_Connection obj_DB_Connection=new DB_Connection();

    Connection connection=obj_DB_Connection.get_connection();

    PreparedStatement ps=null;

    ResultSet rs=null;

    try{

        String query="select * from student";

        ps=connection.prepareStatement(query);

        //ps.setString(1, sl_no);

        System.out.println(ps);

        rs=ps.executeQuery();

        while(rs.next()){

            System.out.println("Sl no -"+rs.getString("sl_no"));

            System.out.println("name -"+rs.getString("name"));

            System.out.println("email -"+rs.getString("email"));

            System.out.println("-----");

        }

    } catch (Exception e) {

        System.out.println(e);

    }

}

```

```

public void update_data(String sl_no,String new_sl_no,String name,String email){

    DB_Connection obj_DB_Connection=new DB_Connection();

    Connection connection=obj_DB_Connection.get_connection();

    PreparedStatement ps=null;

    try{

```

```
sl_no=?";
```

```
String query="update student set sl_no=?,name=?,email=? where
```

```
ps=connection.prepareStatement(query);
```

```
ps.setString(1, new_sl_no);
```

```
ps.setString(2, name);
```

```
ps.setString(3, email);
```

```
ps.setString(4, sl_no);
```

```
System.out.println(ps);
```

```
ps.executeUpdate();
```

```
} catch (Exception e) {
```

```
System.out.println(e);
```

```
}
```

```
}
```

```
public void delete_data(String sl_no){
```

```
DB_Connection obj_DB_Connection=new DB_Connection();
```

```
Connection connection=obj_DB_Connection.get_connection();
```

```
PreparedStatement ps=null;
```

```
try{
```

```
String query="delete from student where sl_no=?";
```

```
ps=connection.prepareStatement(query);
```

```
ps.setString(1, sl_no);
```

```
System.out.println(ps);
```

```
ps.executeUpdate();
```

```
} catch (Exception e) {
```

```
System.out.println(e);
```

```
}
```

```

    }

    public static void main(String[] args) {

        CRUD_Test objTest=new CRUD_Test();

        //insert data

        objTest.create_data("1", "jinu", "jinu@gmail.com");

        objTest.create_data("2", "jawad", "jawad@gmail.com");

        //read data

        objTest.read_data("2");

        //update data

        objTest.update_data("2", "22", "jawad2", "jawad2@gmail");

        //delete data

        objTest.delete_data("1");

    }

}

```

2.1:

```

package CRUD;

import java.sql.Connection;

import java.sql.DriverManager;

public class DB_Connection {

    public Connection get_connection(){

        Connection connection=null;

        try{

            Class.forName("com.mysql.cj.jdbc.Driver");

            connection =

DriverManager.getConnection("jdbc:mysql://localhost:3306/exercise","root", "root123");

```

```

    } catch (Exception e) {

        System.out.println(e);

    }

    return connection;

}

public static void main(String[] args) {

    DB_Connection obj_DB_Connection=new DB_Connection();

    System.out.println(obj_DB_Connection.get_connection());

}
}

```

OUPUT:

The screenshot shows an IDE with two panels. The left panel displays the source code of a Java application, and the right panel shows the console output.

Source Code (Left Panel):

```

53●    try {
54        String query="update student set sl_no=?,name=?,email=? where sl_no=?";
55        ps=connection.prepareStatement(query);
56        ps.setString(1, new_sl_no);
57        ps.setString(2, name);
58        ps.setString(3, email);
59        ps.setString(4, sl_no);
60        System.out.println(ps);
61        ps.executeUpdate();
62●    } catch (Exception e) {
63        System.out.println(e);
64    }
65
66
67●    public void delete_data(String sl_no){
68        DB_Connection obj_DB_Connection=new DB_Connection();
69        Connection connection=obj_DB_Connection.get_connection();
70        PreparedStatement ps=null;
71●    try {
72        String query="delete from student where sl_no=?";
73        ps=connection.prepareStatement(query);
74        ps.setString(1, sl_no);
75        System.out.println(ps);
76        ps.executeUpdate();
77●    } catch (Exception e) {
78        System.out.println(e);
79    }
80
81
82
83●    public static void main(String[] args) {
84        CRUD_Test objTest=new CRUD_Test();
85        //insert data
86        objTest.create_data("1", "jinu", "jinu@gmail.com");
87        objTest.create_data("2", "jawad", "jawad@gmail.com");
88        //read data
89        objTest.read_data("2");
90        //update data
91        objTest.update_data("2", "22", "jawad2", "jawad2@gmail");
92        //delete data
93        objTest.delete_data("1");
94    }
95 }
96
97

```

Console Output (Right Panel):

```

<terminated> CRUD_Test [Java Application] C:\Users\rudra\p2\pool\plugins\org.eclipse.justi.openjdk hotspot.jre
com.mysql.cj.jdbc.ClientPreparedStatement: insert into student(sl_no,name,email) values(1,jinu,jinu@gmail.com)
com.mysql.cj.jdbc.ClientPreparedStatement: select * from student
Sl no -1
name -jinu
email -jinu@gmail.com
-----
Sl no -2
name -jawad
email -jawad@gmail.com
-----
Sl no -22
name -jawad2
email -jawad2@gmail
-----
com.mysql.cj.jdbc.ClientPreparedStatement: update student set sl_no=?,name=?,email=? where sl_no=?
java.sql.SQLIntegrityConstraintViolationException: Duplicate entry '22' for key 'PRIMARY'
com.mysql.cj.jdbc.ClientPreparedStatement: delete from student where sl_no=?

```

